

7.2.3 Motion Capture Controller

The Motion Capture controller uses input from an external device to control the animation of an object in real time. The controller can use a number of different input devices such as a mouse, keyboard, joystick, or a MIDI controller. After the controller is assigned and the input device is configured, the actual motion is captured by using the Motion Capture utility in the Utilities panel. Expressions use mathematical formulas to control animation. Scripts use MAXScript programming to control animation.

7.2.4 Script and Expression Controllers

3ds Max has controllers that allow you to manage the motion of an object by using either expressions or scripts. Expressions are mathematical equations that connect the motion of one object to another. Scripts are more complex and allow for the inclusion of MAXScript, a programming language native to 3ds Max.

7.3 Creating keys

In 3ds Max, animation is defined by keys, which record parameter changes over time. Almost anything in 3ds Max can be animated, from an object's position in space to its shape and colour. There are two main ways of creating keys: Set Key and Auto Key. When either is active, the viewport is bordered in red to let you know that you are key framing.

7.3.1 Working with the Time Slider

The time slider allows you to move quickly from one frame to the next in order to set key frames or scrub through your animation. The slider is a horizontal bar, divided into frames. Time Configuration lets you define the frame rate, time display, real-time playback, and the length of the animation.

- **Keys**

Each key in the timeline shows up as a box at the keyed frame. Each box is colour-coded: red represents a position key, green is rotation, and blue is scale. Grey keys are for non-transform parameters.

- **Slider**

Left clicking and dragging moves the time slider. Sound is imported into 3ds Max. You can then display it by right-clicking in the time slider and choosing sounds from the menu. Visualizing the sound against your keys helps the animation process.

- **Selecting keys**

Left-click and drag to select multiple keys. You can then left-click and